Agent v0.1

2026 3 19 StablePulse Research v0.1 Draft

2025 2026 AI Agent Coinbase x402 Google AP2 Stripe ACP/SPT Tempo MPP agent agent

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Agent

1.1

AI **API Key +**

	agent
Agent	A agent B agent
	Agent API Key

1.2 Agent

- KYC
- HTTP
-
-

2.1 x402 Coinbase

HTTP stablecoin

2025 5 V1 2026 1 V2

Coinbase Developer Platform x402.org Coinbase

HTTP `402 Payment Required` HTTP

```
1. Client 请求资源
2. Server 返回 402 + 支付要求（金额、网络、币种）
3. Client 创建签名 payload
4. Client 携带支付头重新请求
5. Facilitator 验证签名 + 链上余额
6. 提交链上交易，等待确认
7. Server 响应资源 + 结算确认
```

Facilitator

Facilitator

-
-
- Client ERC-20 permit Facilitator

- Base Solana Ethereum L2 EVM
- USDC
- **SDK** TypeScript / Python / Go / Java / Rust
- **V2**
  - Wallet-based Sessions
  - Discovery extension
  - CAIP
  - ACH/SEPA
  - scheme

2026 3

	1 +
	580+
GitHub Stars	5,700+
Facilitator	30+
	TypeScript / Python / Go / Java / Rust

- **Cloudflare** x402 Foundation Agents SDK MCP Server
- **Stripe** 2026 2 Base USDC
- **Google** AP2 A2A x402 x402
- **Coinbase Agentic Wallets** 2026 2 agent

2.2 AP2 — Agent Payments Protocol Google

payment-agnostic

2026 1

Google Cloud 60+

> *AI agent*

VDC

VDC		

Intent Mandate	agent	
Cart Mandate		+
Payment Mandate		" agent "

Intent Mandate

JWT ES256K Secure Enclave/TPM

```
{
  "agent_id": "did:example:agent123",
  "allowed_categories": ["travel"],
  "max_amount": { "value": 500, "currency": "USD" },
  "valid_until": "2026-06-01T00:00:00Z",
  "signature": "用户私钥签名"
}
```

W3C VC JWT DID

60+

- **/** Mastercard American Express PayPal Adyen Worldpay JCB UnionPay Revolut
- **/AI** Coinbase MetaMask Ethereum Foundation Salesforce ServiceNow Intuit
- **/** Etsy Ant International

Google

```
UCP（Universal Commerce Protocol）—— 商品发现与商务层
        ↓ 支付集成
AP2（Agent Payments Protocol）—— 授权与可信层
        ↓ 链上支付子协议
A2A x402 Extension —— 链上执行层（Coinbase + MetaMask + Ethereum Foundation 联合开发）
```

AP2 " " x402

- agent
- Mastercard PayPal
-

2.3 ACP + SPT Stripe + OpenAI

2025 9 ACP 2026 SPT

ACP Agentic Commerce Protocol

AI agent **ChatGPT Instant Checkout**
ChatGPT Etsy/Shopify

SPT Shared Payment Token

agent " agent"

```
1. 买家在 AI 平台保存支付方式
2. 发起购买时，平台生成一次性 SPT（限时/限额/限商家/可撤销）
3. SPT 标识符通过 API 发给商家
4. 商家用 SPT 创建 PaymentIntent
5. Stripe 执行交易 + 欺诈检测 + Token 约束执行
```

- Programmable controls
-
- Stripe Radar
- PCI DSS agent

Microsoft Copilot Anthropic Perplexity Vercel Lovable Replit Bolt Manus

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2.4 MPP — Machine Payments Protocol Tempo / Stripe + Paradigm

M2M

2026 3 18

Tempo Stripe Paradigm 2025 5 50

- stablecoin
-
-

```
┌──────────────────────────────────────────────────┐
│  商务层（买什么、怎么结账）                          │
│  ACP + SPT（Stripe + OpenAI）                      │
│  → 消费者 agent 购物、ChatGPT Instant Checkout      │
├──────────────────────────────────────────────────┤
│  授权层（谁授权、谁负责）                           │
│  AP2（Google + 60 家机构）                         │
│  → 企业级 agent、跨平台授权、加密证据链              │
├──────────────────────────────────────────────────┤
│  执行层（钱怎么从 A 到 B）                          │
│  x402（Coinbase + Cloudflare）                     │
│  → HTTP 原生、stablecoin、开发者/API/crypto-native  │
├──────────────────────────────────────────────────┤
│  基础链（高频 M2M 底层轨道）                        │
```

```
│  MPP / Tempo (Stripe + Paradigm)                          │
│  → 专用链，法币+加密双轨，刚上线                              │
└──────────────────────────────────────────────────────────┘
```

Google AP2 x402 A2A x402 Extension

4.1

agent API Key

Web3

- Coinbase Agentic Wallets 2026 2 agent x402
- Alchemy 2026 3 agent HTTP 402 Base USDC

Web2

- **Ramp Agent Cards** 2026 3 11 agent Ramp API Visa token

4.2 Agent

orchestrator agent agent

" " A2A

- Agoragentic 26+ REST
- Questflow agent
- x402 agent

agent

4.3

agent

x402

- Mastercard Agent Pay 2025 2026 3 Santander agent
- Visa Intelligent Commerce VisaNet AI-Ready Cards

agent

Agent

5.1

	AP2 Intent Mandate	+		Web2 + Web3	

	Safe +				
ZK	zkTLS zkP2P				ZK

5.2

```
[用户意图层]  AP2 Intent Mandate（JWT 签名，定义边界）
      ↓
[执行控制层]  智能合约钱包（Safe，链上强制执行）
      ↓
[结算层]      x402（HTTP 内 USDC 支付）
      ↓
[服务证明层]  zkTLS（证明服务确实被提供，防止欺诈）
```

OpenClaw

OpenClaw AI Agent

	API Key + Web2
	ClawHub 13,700+ 311+
	CoinFello + MetaMask ERC-4337 + ERC-7710
	CZ 2026 3 AI agent
	2026 2 OpenAI

skill " " " "

7.1

Crypto-native agent DeFi	x402
agent SaaS B2B	AP2 + Stripe ACP
agent	ACP + SPT Stripe + OpenAI
M2M	MPP Tempo x402 V2

7.2 x402 Coinbase

x402 Coinbase

- Base agent
- USDC agent
- CDP agent

7.3

```
2025        2026            2027            2028+
  |           |               |               |
x402 发布    生态快速扩张      垂直场景引爆?     基础设施化?
AP2/ACP发布  Web2/Web3并行     标准竞争白热化     协议分层稳定
```

—— AI API agent marketplace

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1. **x402 agent** 1 +
2. **AP2**
3. **Web2 Ramp Mastercard Visa Web3**
4. AP2 x402
5. **Agent autonomous agent**
6. Web2 Ramp Web3 x402

- x402 GitHub Repository
- x402 Ecosystem Directory
- x402 V2 Launch Blog
- AP2 Protocol Documentation
- Launching the x402 Foundation | Cloudflare Blog
- Introducing Agentic Commerce Solutions | Stripe
- Announcing AP2 | Google Cloud Blog
- Stripe-led Tempo goes live with MPP | CoinDesk
- Ramp Launches Agent Cards
- Agentic Payments Explained: ACP, AP2, x402 | Orium
- CoinFello Launches OpenClaw Skill | DL News
- 31 AI Agent Payment Statistics | Nevermined

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v0.1 | 2026-03-19 | v0.2 MPP Visa Intelligent Commerce